

ATVB IN FOCUS:
RNA Modifications in Cardiovascular Disease

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Adenosine-to-Inosine (A-to-I) RNA Editing by ADAR1 to Control RNA Sensing in Cardiovascular Disease

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ABSTRACT: Across biology, organisms have retained a mechanism to diversify the RNA transcriptome through RNA editing. Mediated by ADAR (adenosine deaminase acting on RNA) enzymes, Adenosines in double-stranded RNA (dsRNA) structures can be edited to Inosines (adenosine-to-inosine edit). Although this can change the amino acid sequence if it occurs in a coding sequence of mRNA, the majority of RNA editing in mammalian cells is found in noncoding repetitive elements. These repetitive elements have a predisposition to form long dsRNA structures that can mimic a dsRNA virus. Since the initial discoveries of RNA editing over 30 years ago, investigators have now identified ADAR1 to play a crucial role in suppressing innate immune activation and type I interferon signaling. Through adenosine-to-inosine editing, these dsRNA change their conformational structures and evade activation of the innate immune dsRNA sensor, MDA5 (melanoma differentiation-associated gene 5; gene symbol *IFIH1*). In human disease, although rare loss-of-function variants of ADAR1 have been associated with severe autoimmune disease, there has also been a rapid advance in our understanding of this molecular pathway in common complex diseases. We now understand that common genetic variants can impact RNA editing frequencies, and variants that decrease RNA editing are associated with an increase in risk of numerous autoinflammatory disorders as well as coronary artery disease. This rapid advance in our understanding of the genetic determinants of RNA editing and coronary artery disease has been mirrored by new discoveries in molecular biology, where deficient RNA editing within the vascular wall and smooth muscle cell now highlights endogenous RNA sensing by MDA5 as a causal mechanism of coronary artery disease and other vascular disorders. Here, in this review, we provide a focused look at major advances in RNA editing and cardiovascular disease and put these discoveries into historical context with a goal to map the next steps to advance these molecular pathways to new therapeutic discovery.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: adenosine deaminase ■ cardiovascular diseases ■ coronary artery disease ■ RNA, double-stranded ■ RNA editing

Although the central dogma dictates that the sequence of DNA ultimately regulates RNA transcription to guide protein structure and function, organisms have evolved a distinct capacity to diversify the RNA transcriptome by modifying RNA sequence itself.¹ In this mechanism, mediated by a class of enzymes called ADARs (adenosine deaminase acting on RNA), RNA can undergo adenosine-to-inosine (A-to-I) RNA editing, a chemical deamination step that functionally results in an adenosine-to-guanosine modification. In a fascinating aspect of biology, the ability for cells to regulate RNA sequence through ADAR RNA editing is highly conserved.¹ ADAR RNA editing is found

in invertebrate species, such as coral, cephalopod species, such as octopus and squid, as well as mammalian species, including rodents, nonhuman primates, and humans.¹ Although the function of ADAR RNA editing has distinct differences across species, the importance of this mechanism is clear, highlighting a unique question in evolution as to the selective pressures that governed this unique biological adaptation.²

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Nonstandard Abbreviations and Acronyms

ADAR	adenosine deaminase acting on RNA
AGS	Aicardi-Goutières Syndrome
A-to-I	adenosine-to-inosine
CAD	coronary artery disease
dsRNA	double-stranded RNA
edQTL	editing quantitative trait loci
eNOS	endothelial NO synthase
IFN	interferon
IL	interleukin
ISG	interferon-stimulated genes
KD	knock-down
LOF	loss-of-function
NFκB	nuclear factor kappa B
PAH	pulmonary arterial hypertension
SMC	smooth muscle cell
TF	transcription factor

FROM RNA SEQUENCE TO RNA STRUCTURE

RNA editing was discovered in the late 1980s^{3,4}; however, our understanding of its biological function as well as its relevance to human disease has continued to undergo significant change (recently reviewed in Li and Walkley⁵). The modification of an RNA sequence with A-to-I (G) editing can change the amino acid sequence and ultimately protein structure and function when occurring in a coding sequence of an RNA transcript. An amazing example of this was recently demonstrated in the octopus, where in response to a change in ocean temperature, the octopus will rapidly edit thousands of RNAs to diversify its transcriptome to change the neuronal proteome.⁶

In mammals, it was discovered that specific tissues appeared to have distinct RNA editing frequencies.⁷ From the analysis of thousands of human samples as well as hundreds of other primate and mouse samples, it was identified that the brain and arteries have the highest frequency of RNA editing.⁷ An important observation made in the early 2000s was that A-to-I RNA editing across the human transcriptome is highly frequent within the *Alu*-containing noncoding regions of mRNA.⁸ This raises an important consideration that while RNA editing of coding sequences can affect protein structure, the vast majority of RNA editing sites in mammalian transcriptomes are in noncoding repetitive short interspersed nuclear elements (ie, *Alu* repeats in human, *B1/B2* repeats in mouse) that, through RNA folding, have the capacity to form double-stranded RNA (dsRNA) structures.

There are 3 human ADAR genes (*ADAR*, *ADARB1*, and *ADARB2*) that respectively code for ADAR1, ADAR2, and ADAR3 proteins. These proteins have a strong preference

Highlights

- Discovered over 30 years ago, adenosine-to-inosine RNA editing by ADAR (adenosine deaminase acting on RNA) enzymes has evolved from a biochemical curiosity to a central regulator of transcriptomic diversity and innate immune tolerance.
- ADAR1-mediated editing of endogenous double-stranded RNA prevents inappropriate activation of the innate immune sensor MDA5 by altering immunogenic RNA structure.
- Although rare loss-of-function variants in ADAR1 cause severe Mendelian interferonopathies, common genetic variants that reduce RNA editing (editing quantitative trait loci) increase the risk of inflammatory disorders, including coronary artery disease.
- Vascular smooth muscle cells exhibit a unique requirement for ADAR1, where impaired editing activates MDA5 and accelerates atherosclerotic plaque progression and calcification.
- Human genetic, transcriptomic, and functional experimental data converge to establish the ADAR1–double-stranded RNA–MDA5 axis as a causal mechanism in atherosclerosis and are a promising therapeutic target.

to bind to dsRNA where their deaminase domain can facilitate A-to-I editing at selected RNA editing sites. Across the mammalian transcriptome, it was discovered that ADAR1 does the bulk of RNA editing and primarily edits noncoding regions of the mRNA transcript, while ADAR2 does a minority of RNA editing, and notably regulates the editing of coding sequences.⁷ ADAR3 lacks a deaminase domain, and although it can bind to RNA, it does not perform editing itself and is only expressed in selected tissues.⁵

The high frequency of RNA editing in arteries raises a unique question as to whether RNA editing has a specific role in the pathogenesis of vascular diseases. In 2012, it was reported that rare variants that affect the deaminase domain of ADAR1 (*ADAR*) were associated with a severe pediatric autoimmune disorder, Aicardi-Goutières Syndrome (AGS), classified as a type I Interferonopathy.⁹ The authors discovered that through loss of RNA editing capacity, AGS patients were characterized as having distinct gene signatures of IFN (interferon)-stimulated genes (ISGs) in whole blood samples. Importantly, in ADAR1-associated AGS, patients have unique pathology, including abnormal brain development, as well as vascular pathology, such as vascular inflammation and calcification,⁹ fitting with what was later described as 2 sites with unusually high RNA editing frequency.

Only a few years before this discovery of ADAR1 in the pathogenesis of AGS, it was identified that ADAR1 played a critical role in suppressing IFN signaling.¹⁰ However, the

mechanism by which ADAR1 and RNA editing may mediate this effect was not clear. One year later, in 2010, it was discovered that the presence of inosines through IU pairs in dsRNA suppresses ISG activation.¹¹ It was then recognized that the long dsRNA structures in unedited RNAs can mimic a dsRNA virus and activate an innate immune response through activating dsRNA sensors (see schematic diagram Figure 1). This created a new conceptual model that, through ADAR1 A-to-I editing, dsRNA may be sensed as self and evade innate immune RNA sensing from cytosolic RNA sensors; however, if unedited, dsRNA is sensed as nonself and activates an intrinsic innate immune response.¹¹ ADAR1 has 2 major isoforms, where ADAR1p150 is notably inducible by IFN α and largely resides in the cytoplasm, while ADAR1p110 resides in the nucleus (Figure 1).⁵ ADAR1p150 function is distinct in that it is primarily responsible for editing of immunogenic RNA that suppresses ISG activation.¹²

These discoveries created a hypothesis that ADAR1 RNA editing can suppress ISG activation through changing dsRNA structure and prevent dsRNA sensing and innate immune activation. Although embryonic lethality in mice with *Adar* deletion was previously described,^{10,13} a reasonable hypothesis may be that if one can eliminate ISG activation by preventing dsRNA sensing, the loss of RNA editing itself may have minimal biological effect. To test this, investigators developed a mouse model that recapitulates the loss of deaminase function in human AGS patients, where homozygous mice are embryonic lethal, but with the codeletion of the long dsRNA sensor

MDA5 (gene symbol *Ifih1*, interferon induced helicase C domain 1), they lived until adulthood.¹⁴ An identical observation was made at nearly the same time in 2015, which further revealed MDA5 signaling through MAVS (mitochondrial antiviral signaling protein) to be essential in mediating downstream ISG signaling and mortality.¹⁵ These observations were critical in defining a concept that the function of ADAR1 RNA editing is through incorporation of inosines into long repetitive dsRNA elements, modifying the structure of hundreds of immunogenic dsRNAs to eliminate endogenous dsRNA sensing by MDA5 (schematic diagram, Figure 1). We now understand that ADAR1 evades MDA5 activation through the editing of several hundred immunogenic RNAs with a particular emphasis on a relatively small subset.¹² Prior work has also revealed that after activation of MDA5, apoptosis and cell cycle arrest are driven in part by activation of protein kinase R, a protein that is a key regulator of cellular function and transcriptional control after viral infection.¹⁶

This is in contrast to ADAR2, where deletion of *Adar2* (*Adarb1*) in mouse results in perinatal mortality due to seizures from lost editing of the AMPA (α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid) receptor *Gria2*.¹⁷ In this model, investigators revealed that lethality can be prevented through introducing a point mutation to recapitulate the edited site in the *Gria2* gene, implicating a primary importance of *Adar2* in editing of this single specific transcript.¹⁷ However, there is recent evidence that supports endothelial cell-specific ADAR2 to have key roles in regulating immune cell trafficking and

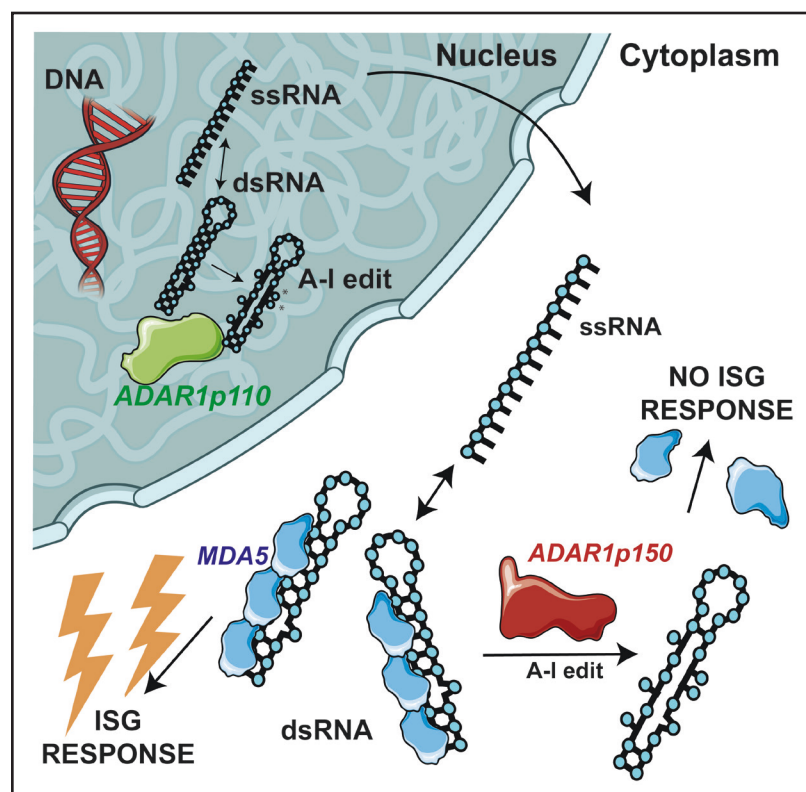


Figure 1. ADAR1 (adenosine deaminase acting on RNA 1) RNA editing of double-stranded RNA (dsRNA) is required to avoid dsRNA activation of MDA5 (melanoma differentiation-associated gene 5; gene symbol *IFIH1*) and deleterious interferon-stimulated gene (ISG) transcriptional response.

Schematic diagram showing single-stranded RNA (ssRNA) transcription with dsRNA formation and adenosine-to-inosine (A-to-I) RNA editing by ADAR1p110 in the nucleus, and then ssRNA and dsRNA formation in the cytoplasm and A-to-I RNA editing by ADAR1p150 in the cytoplasm. ADAR1 RNA editing of dsRNA alters dsRNA structure and evades MDA5 sensing and ISG response.

inflammatory response to IL (interleukin) 6, with likely important implications in disease.¹⁸

BEYOND RARE DISEASE: ADAR1 IN ATHEROSCLEROSIS

Although our understanding of the role of ADAR1 to regulate dsRNA structure and dsRNA sensing by MDA5 in rare diseases such as AGS was further clarified in 2015, it was shortly after this time that investigators proposed ADAR1 to have distinct roles in complex disorders such as atherosclerosis. Investigators had identified that ADAR1 has specific roles to edit a hyper-editable RNA transcript *CTSS*, and it was suggested that through ADAR1 editing of the 3' UTR (untranslated region) of *CTSS*, expression of *CTSS* is regulated by modifying the RNA binding protein HuR,¹⁹ thus impacting endothelial cell inflammation and disease risk. This report had clearly identified that RNA editing occurs in human atherosclerosis and implicated an intriguing mechanism; however, this mechanism was perhaps in contrast to that implied by the recent reports suggesting the dominant effect from lost ADAR1 RNA editing to be mediated by dsRNA sensing. Similarly, these investigators further proposed ADAR1 to edit alternative RNAs, such as the lncRNA (long non-coding RNA) *NEAT1*, and that *NEAT1* editing affects its stability and function with implications in atherosclerotic disease risk.²⁰

Simultaneously, investigators identified the vascular smooth muscle cell (SMC) to have distinct requirements for ADAR1 RNA editing, where conditional SMC-specific deletion of *Adar* results in lethality and SMC-specific haploinsufficiency of *Adar* regulates SMC phenotype in

models of carotid injury and aneurysm.^{21,22} However, the mechanisms proposed in these reports were focused on the potential for ADAR1 editing of specific transcripts to influence their function, including the modification of RNA splicing, and were studied outside the context of dsRNA sensing. These important studies highlighted a key observation—RNA editing occurs in atherosclerosis, ADAR1 appears to be driving the dominant portion of RNA editing, and ADAR1 RNA editing has key roles in the vascular wall and SMC phenotype.

ADAR1-dsRNA-MDA5 AXIS IN ATHEROSCLEROSIS

Although impaired RNA editing and increased dsRNA sensing were implicated in rare diseases, such as AGS and subsequently a related disorder Singleton-Merton Syndrome,²³ a key discovery in 2022 suggested a strong genetic signal of the ADAR1-dsRNA-MDA5 axis in the pathogenesis of complex autoinflammatory disorders, including atherosclerosis.²⁴ Although rare loss of deaminase function of ADAR1 leads to broad impairments in RNA editing across the transcriptome and a type I interferonopathy,⁹ through analysis of thousands of human RNA sequencing data sets combined with genotype data through GTEx (gene tissue expression project), investigators were able to reveal that common genetic variants can influence the frequency of RNA editing (schematic diagram, Figure 2).²⁴ These common single nucleotide polymorphism variants, termed editing quantitative trait loci (edQTL), carried a significant burden of risk for numerous autoinflammatory disorders, where edQTLs decreasing RNA editing frequency increased

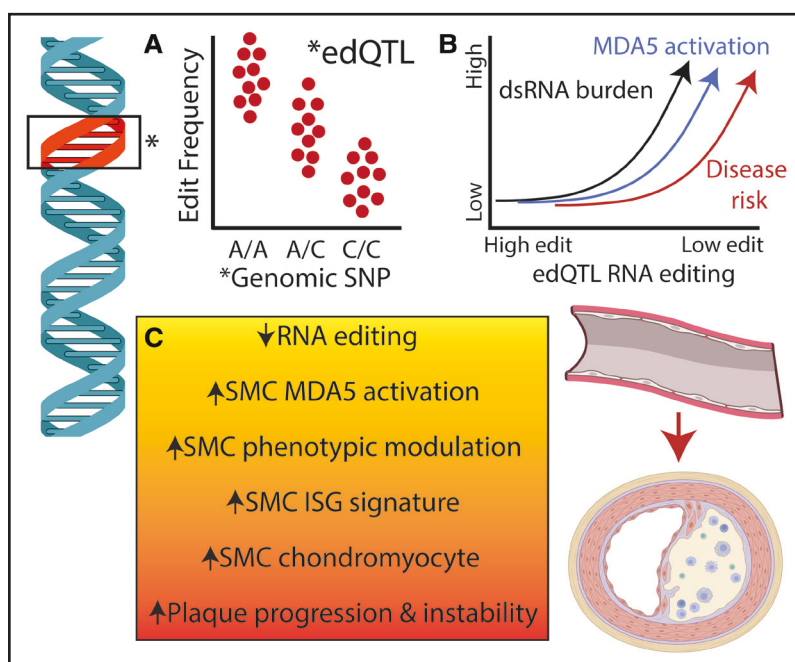


Figure 2. Common genetic variants can modify RNA editing, influence double-stranded RNA (dsRNA) formation, MDA5 activation, and disease risk.

A, Schematic diagram representing common single nucleotide polymorphisms (SNPs) that modify RNA editing frequency, termed editing quantitative trait loci (edQTLs). **B**, Decreased RNA editing from edQTLs is associated with increased dsRNA burden, increased MDA5 activation, and increased disease risk for numerous common autoinflammatory disorders, including psoriasis, lupus, type 1 diabetes, as well as coronary artery disease. **C**, Decreased RNA editing in the vascular wall leads to smooth muscle cell (SMC)-specific activation of MDA5, altered SMC phenotypic state, increased SMC interferon-stimulated genes (ISG) signature, increased SMC chondromyocyte formation, and vascular calcification, and increased plaque progression and instability.

the risk of not only disorders including psoriasis, type 1 diabetes, inflammatory bowel disease, rheumatoid arthritis but also coronary artery disease (CAD; Figure 2).²⁴ Intriguingly, the authors identified that roughly 20% of CAD risk genome-wide association study loci colocalized with edQTL loci, and that when predicting variant effect on disease heritability, edQTLs had a greater effect on disease heritability of numerous complex disorders over traditional expression QTLs.²⁴

When evaluating individual tissue-level expression of ISGs and edQTLs in coronary arteries, the authors confirmed that decreasing RNA editing (via edQTLs) was associated with increasing ISG signatures, further connecting this axis of ADAR1-dsRNA-MDA5 in disease pathogenesis. This relationship was similarly shown using synovial tissues for patients with rheumatoid arthritis, brain white matter tissue for patients with multiple sclerosis, and peripheral blood mononuclear cell samples from patients with lupus.²⁴

This work established an axis of ADAR1-dsRNA-MDA5 as a genetic mechanism of common complex inflammatory diseases, and the relationship to CAD is further supported by the notable discovery that loss-of-function (LOF) variants in *IFIH1* (MDA5) are protective against CAD²⁵ and that the *IFIH1* and *ADAR* loci meet genome-wide level significance for association to CAD in a genome-wide association study.²⁶ Although the exact threshold where edQTLs and unedited immunogenic RNAs can trigger MDA5 is not well defined, these data implicate hundreds of edQTLs and RNAs that collectively lead to MDA5 activation and contribute to disease. However, while the genetic evidence for ADAR1 RNA editing in atherosclerosis and vascular biology implicates dsRNA sensing by MDA5, this mechanism was in contrast to prior reports where emphasis had been placed on ADAR1 editing of specific transcripts influencing individual transcript function.^{19–22} Similarly, the mechanism by which ADAR1-dsRNA-MDA5 may influence disease risk was not established and was a key gap in our knowledge.

LINKING ADAR1-dsRNA-MDA5 AS A CAUSAL MECHANISM OF CAD

In our recent work,²⁷ we built on this strong human genetic evidence that supports the ADAR1-dsRNA-MDA5 axis to play an important role in the pathogenesis of atherosclerosis (schematic diagram, Figure 2). Through human single-cell RNA sequencing, culture of primary human coronary artery SMCs, mouse models of SMC-specific deletion of *Adar* in homozygous studies as well as in haploinsufficient 16 week atherosclerosis studies with additional dual deletion of *Ifih1* (*Mda5*), combined with evaluation of a large human carotid atherosclerosis genomics database (AtheroExpress), we established a

series of principles that supports a paradigm whereby impaired SMC-specific ADAR1 RNA editing activates MDA5 in atherosclerosis and through MDA5 activation there are distinct alterations in SMC phenotype that promotes increased plaque development, plaque calcification, and disease pathogenesis.²⁷

Loss of SMC-Specific ADAR1 Regulates SMC Phenotype via MDA5

Although prior reports implicated impaired SMC *Adar1* to influence SMC phenotype,^{21,22} these reports had not studied SMC phenotypic change in relation to RNA sensing. In our recent report, we reveal that through culture of primary human coronary artery SMCs, knock-down (KD) of *ADAR* results in marked reduction in global RNA editing, and this is associated with downregulation of mature SMC contractile markers while there is notable upregulation of key ISGs as well as important TFs (transcription factors), such as *KLF4*, *EGR1*, and *ATF3*.²⁷ Regulation of key developmental TFs such as these are critical in mediating SMC phenotype and progression of atherosclerosis (recently reviewed in Serbulea et al²⁸), and although activation of MDA5 is known to signal through MAVS and subsequently downstream factors including TBK1 (TANK binding kinase 1) and IRF3 (interferon regulatory factor 3),²⁹ the specific mechanism by which MDA5 can regulate TFs, such as *KLF4* (Kruppel like factor 4), *EGR1* (early growth response 1), and *ATF3* (activating transcription factor 3), is not well mapped out. However, we further showed through models of induced phenotypic transition with serum stimulation, that the effect from *ADAR* KD on ISG induction and expression of key TFs is enhanced with phenotypic modulation, suggesting an interaction between not only loss of *ADAR* in mediating cellular phenotype, but cellular phenotype itself can modify the effect of *ADAR* loss. Critically, the effect of *ADAR* KD is entirely lost with co-KD of *IFIH1* (*Mda5*). This highlights that the effect of *ADAR* KD on SMC phenotype and ISG activation is entirely dependent on MDA5 and dsRNA sensing. This finding may suggest that if dsRNA sensing is eliminated, the biological effect of *ADAR* deletion is somewhat minimal in human SMCs.

SMCs Have Distinct Requirement for ADAR1 RNA Editing

A specific function of ADAR1 is to bind dsRNA and facilitate A-to-I editing at specific sites, and given ADAR1 can edit thousands of RNA transcripts, a question may be, what dsRNAs are particularly responsible for activation of MDA5? Through a series of experimentation and computational approaches, investigators have

largely identified a series of ≈600 to 700 immunogenic RNA, and highlight specific RNAs (≈150) that are notably immunogenic.¹² Through analysis of a high-quality human single-cell RNA sequencing data set collected from carotid plaque, we identified that cell types have specific expression patterns of these immunogenic RNAs.²⁷ Notably, we revealed that the vascular SMC has considerably higher expression of these immunogenic RNAs compared with other cell types. We then identified markers of ISG activation with SMC phenotypic modulation in atherosclerosis, perhaps suggesting MDA5 activation. To better characterize the role of SMC-specific ADAR1 in vivo, we generated a conditional SMC-specific *Adar* deletion model in mice (*Adar*^{flox/flox}, *Myh11*^{CreERT2}, *ApoE*^{-/-}, *ROSA*^{tdTomato}), where we further revealed that on treatment with tamoxifen to induce homozygous deletion of *Adar* in *Myh11*-derived SMCs, mice become severely ill within days and show >50% mortality by 2 weeks.²⁷ Aortic tissue shows severe pathology, including intramural hemorrhage and loss of vascular integrity. Single-cell RNA sequencing demonstrates not only distinct ISG activation, but clear immune cell recruitment with notable macrophage infiltration and evidence of SMC phenotypic modulation with downregulation of mature SMC contractile markers.²⁷

The reason why the vascular SMC has a notable increase in expression of immunogenic dsRNA is unclear and is an area of active investigation. The immunogenic RNA does not have enrichment for specific biological pathways; however, we do reveal that expression of these collective immunogenic RNA appear to be negatively regulated by activation of MDA5 itself, suggesting the potential for dynamic control of expression.²⁷ Similarly, the specific immunogenic RNAs that may confer a higher degree of MDA5 activation are not well characterized. Although we reported several RNAs to colocalize with edQTL and CAD genome-wide association study and were differentially expressed in our models of SMC phenotypic transition (including *SWAP70* and *TCTA*), the exact relative importance has yet to be determined.²⁷

Loss of SMC *Adar1* Is Lethal Due to Activation of MDA5 in Mouse

As we had shown from in vitro studies that the effect of ADAR KD in SMCs is due to activation of MDA5, we then bred an additional allele of constitutive *Ifih1*^{-/-} (Mda5) onto the SMC *Adar* KO mouse, and revealed that when Mda5 is deleted, the effect of SMC *Adar* KO is entirely prevented. Importantly, the effect of SMC *Adar* KO was entirely prevented with simply *Ifih1*^{-/+}, suggesting haploinsufficiency of Mda5 is sufficient to eliminate endogenous RNA sensing. Impressively, single-cell RNA sequencing of each group (control mice, SMC *Adar* KO, SMC *Adar* KO + *Ifih1* KO) showed that the profound transcriptional changes that occur with SMC *Adar* KO

are entirely prevented with deletion of *Ifih1* (Mda5), and the effects on SMC modulation, such as downregulation of mature contractile markers, were reversed when Mda5 was deleted. This demonstrates a consistent finding from our in vitro data that in the SMC, ADAR1 plays a critical role in editing immunogenic RNA to evade dsRNA sensing by MDA5, and it is through activation of MDA5 that regulates a complex transcriptional network that influences ISG activation, SMC phenotypic change, and cell-cell communication and immune cell recruitment.

Haploinsufficiency of SMC *Adar1* Accelerates Atherosclerosis via MDA5

While in early studies it was suggested that haploinsufficiency of ADAR1 in mice is well tolerated without any phenotype, prior work implicated that haploinsufficiency of ADAR1 in SMCs influences SMC phenotype in specific models of carotid injury and aneurysm.^{21,22} This perhaps suggests a relationship between the demands for ADAR1 RNA editing and the kinetic activity of the ADAR1 protein, which may be cell type and cellular context-specific. Given the severity of phenotype in the SMC *Adar* KO mouse, we investigated how SMC *Adar* haploinsufficiency may affect atherosclerosis progression in a hyperlipidemia model of atherosclerosis (*ApoE*^{-/-} with 16 weeks of high-fat diet) with additional SMC lineage tracing (*Adar*^{flox/het}, *Myh11*^{CreERT2}, *ApoE*^{-/-}, *ROSA*^{tdTomato}). With tamoxifen treatment at 8 weeks of age, we revealed that SMC *Adar* haploinsufficiency leads to specific ISG activation in SMCs undergoing phenotypic modulation, and that activation of these pathways leads to altered SMC phenotypic trajectory, increasing vascular chondromyocyte formation—a critical cell type for the formation of plaque intimal calcification. These mice had increased plaque size, marked ISG expression throughout plaque with a preference for the plaque base, increased SMC-derived cell engraftment into the plaque with decreased medial wall thickness, and a notable increase in calcification by staining for alkaline phosphatase.²⁷ Importantly, we then repeated these experiments with the incorporation of an *Ifih1*^{-/+} (Mda5) allele where we could test if this effect from SMC *Adar* haploinsufficiency is dependent on Mda5. Significantly, the effect of SMC *Adar* haploinsufficiency was entirely prevented with concurrent haploinsufficiency of *Ifih1* (Mda5). This proved the key relationship between inadequate ADAR1, increased dsRNA formation in a cell type and context-specific mechanism, leading to activation of MDA5 that through regulating a complex network of transcriptional activation and altered cellular trajectory, can increase the development of atherosclerosis. Our data further supported that SMC *Adar* heterozygous + *Ifih1* heterozygous (Mda5) had lower phenotypic modulation and formation of CMCs than that seen in control mice, suggesting Mda5 plays a role in sensing immunogenic RNA even with genetically intact ADAR1. This

data strongly supports the human genetics evidence that implicates the ADAR1-dsRNA-MDA5 axis as a causal mechanism of CAD.

Human Data Corroborates ISG Activation and the Importance of MDA5 in Atherosclerosis

To better understand the link between ADAR1-dsRNA-MDA5 in human atherosclerosis, we utilized the unique AtheroExpress data set that incorporates bulk RNA sequencing, genotype data, and careful plaque histological characteristics from over 1000 carotid endarterectomy samples.²⁷ Our first question was whether the expression of key ISGs is related to markers of SMC phenotypic transition (ie, expression of *LUM* and *HAPLN*). We observed a striking linear relationship between expression of ISGs and expression of *LUM* and *HAPLN*, consistent with our hypothesis that ISG activation occurs with SMC phenotypic modulation. Furthermore, we identified that expression of *IFIH1* (MDA5) itself had an extraordinarily strong relationship to plaque vulnerability ($P=9.71 \times 10^{-135}$) and calcification.²⁷

RAPIDLY CHANGING LANDSCAPE OF ADAR1-dsRNA-MDA5 IN VASCULAR DISEASE

Our work highlights the crucial role of ADAR1 in the vascular SMC to edit immunogenic RNA to evade MDA5 activation. We show that inadequate ADAR1 RNA editing leads to activation of MDA5, and this controls a complex network of thousands of genes that regulates distinct immune responses, including ISG response, cellular phenotype, and cellular fate trajectory.²⁷ We show that impaired SMC *Adar* leads to MDA5 activation in atherosclerosis, and SMC MDA5 activation increases SMC engraftment into plaque, and increases plaque size and calcification through regulating SMC phenotypic trajectory—findings corroborated in large human data sets and supported by human genetics.

Ongoing questions remain: is MDA5 activation only seen with genetically impaired ADAR1? Our human genetics data suggest that common single nucleotide polymorphism variants can influence RNA editing (edQTLs), and a collection of small-effect-size edQTLs increases disease risk for numerous common autoimmune-inflammatory disorders.²⁴ This finding may suggest a gradient effect of dsRNA burden, where increasing dsRNA burden in patients may have a similar increase in burden of MDA5 activation and role in disease. Our human atherosclerosis data from AtheroExpress implicates MDA5 to have a strong relationship to plaque phenotype, such as vulnerability and calcification, despite its investigation in a general carotid atherosclerosis population without specific enrichment for those with high dsRNA burden.

This may suggest MDA5 activation to be a broader mechanism of disease.²⁷ Similarly, our data suggests that MDA5 activation occurs in SMCs in a cellular context-specific mechanism, despite minimal changes in ADAR1 expression. This may reveal a role for dynamic control of ADAR1 enzymatic kinetic function to play a part in regulating MDA5 activation beyond the expression of ADAR1.

A very intriguing recent report has discovered that nuclear eNOS (endothelial NO synthase) interacts with and S-nitrosates ADAR1.³⁰ In this report, investigators revealed that with endothelial dysfunction and loss of NO, endothelial dsRNA accumulates due to impaired ADAR1 function, leading to ISG induction.³⁰ This data further supports that NO is directly interacting with medial/SMC-derived ADAR1, where in a mouse model of endothelial dysfunction, there is increased dsRNA accumulation in the medial wall of coronary arteries. This may explain our observation of increasing ISG induction in SMCs with SMC phenotypic modulation and plaque development, an effect seen with concurrent impairment in endothelial function. This may suggest an interaction between endothelial function and SMC-specific ADAR1 and dsRNA formation.

The importance of SMC-specific ADAR1 was also recently reported in the context of pulmonary arterial hypertension (PAH), where loss of SMC-specific ADAR1 driven by an *Acta2* conditional Cre in mouse exacerbates hypoxemia-induced PAH due to ISG activation.³¹ Similarly, in human PAH, SMCs have distinct activation of ISG signals.³¹ In PAH, investigators also revealed a role for ADAR1 in pulmonary endothelial cells, where loss of ADAR1 signals through MDA5, nocturin (*NOCT*), and IRF7 (interferon regulatory factor 7) to mediate a type I IFN response and PAH pathology.³² The broader role of ADAR1 and its regulation of ISG signaling in cardiovascular disease has also been reported in cardiomyopathy, where cardiomyocyte-specific deletion of *Adar* leads to a profound cardiomyopathy over a period of months after conditional deletion of *Adar*, an effect driven by downstream IRF7 and ISG signals, consistent with MDA5 activation.³³ Human genetics also supports ADAR1-dsRNA-MDA5 axis in aneurysm, where a recent large genome-wide association study of abdominal aortic aneurysm identified the *IFIH1* locus to have a strong association with abdominal aortic aneurysm.³⁴

INHIBITION OF ENDOGENOUS RNA SENSING AS A CLINICAL THERAPY

The collection of human genetic evidence from both rare and complex disease in combination with strong molecular mechanisms of disease in atherosclerosis through SMC-specific MDA5 activation, supports the

ADAR1-dsRNA-MDA5 axis as a causal mechanism of disease. Nonlipid residual risk in atherosclerosis drives a major portion of disease risk, and specific patient populations, including those with underlying autoimmune conditions, show an alarmingly high risk of atherosclerotic CAD, despite control of traditional lipid risk factors.³⁵ This raises the intriguing possibility that inhibition of endogenous RNA sensing through inhibition of MDA5 may be a viable therapeutic strategy.

Given that MDA5 is a viral RNA sensor and plays important roles in mediating innate immune response to viral infection, it may be a reasonable concern that inhibition of MDA5 would pose an unavoidable risk of viral infection and preclude inhibition of MDA5 for any drug development program. However, there are several pieces of data that suggest partial inhibition of MDA5 would be effective and safe. First, Innate immune activation from viral RNA is mediated by multiple factors, including RIG-I (retinoic acid inducible gene 1; gene symbol *DDX58*),³⁶ a key cytoplasmic helicase that senses dsRNA, as well as TLR3 (toll-like receptor 3),³⁷ which activates NF κ B (nuclear factor kappa B) in response to dsRNA viral elements. Foundational work has revealed that cytoplasmic RNA sensors MDA5 and RIG-I have distinct and largely separate roles in mounting this innate immune response to viral RNA. RIG-I senses many viral elements, including rhabdoviruses, paramyxoviruses, orthomyxoviruses (including influenza), flaviviruses, filoviruses, as well as EBV (Epstein-Barr virus) and others. In contrast, MDA5 senses only very specific viral RNAs, such as Picornaviruses, and has partial roles in Reoviruses and Flaviviruses, such as the Dengue virus.^{36,38} This contrast in RNA structural pattern sensing likely reflects why long endogenous dsRNAs can be sensed by MDA5 but not by RIG-I, as RIG-I has a preference for sensing short dsRNA structures. It is understood that RIG-I plays a much larger role in mounting an immune response to viral infections. Importantly, even in viral infections that require MDA5, such as encephalomyocarditis virus, haploinsufficiency of MDA5 in the mouse does not impair the IFN response and protects against viral infection similar to wild-type.³⁶ This is in contrast to our data, where haploinsufficiency of MDA5 is entirely sufficient to prevent endogenous dsRNA sensing.²⁷ This highlights a distinct window where partial inhibition of MDA5 may abrogate adverse sensing of endogenous dsRNA, while allowing for appropriate IFN response to pathogenic viral infections.

Second, predicted LOF (pLOF) variants in MDA5 (*IFIH1*) have been previously identified in exome analysis in the UK Biobank population, where these variants were found to not only be protective against CAD but also other autoinflammatory disorders with no signal of harm.²⁵ Third, we can evaluate how evidence of genetic constraint (where the observed LOF variant frequency falls below what is expected)³⁹ can provide evidence to suggest LOF is not tolerated in a population. Through

gnomADv4, which provides a genomic mutational constraint map in 76 156 human genomes and 730 947 exomes,⁴⁰ MDA5 (*IFIH1*) importantly has no evidence of genetic constraint, where the observed to expected pLOF ratio is 1.17, nearing statistical significance to suggest that pLOF variants are selected for in the population, rather than selected against. This contrasts with RIG-I (*DDX58*), where the observed to expected pLOF ratio is 0.81, or ADAR1 (*ADAR*) with observed to expected=0.35, meeting statistical significance for genetic constraint.

Although rare homozygous LOF or dominant negative heterozygous variants of *IFIH1* (MDA5) have been associated with increased risk of infection,^{41,42} parents of these individuals with heterozygous LOF are unaffected, and these data are consistent that there is a different threshold for endogenous RNA sensing compared with viral RNA sensing from MDA5. Data support that partial inhibition of MDA5 can eliminate endogenous dsRNA activation of MDA5 while retaining appropriate viral RNA sensing. Importantly, inhibition of MDA5 represents a highly selective mechanism of inhibition of endogenous RNA sensing while not impacting any other aspect of innate immune response. This contrasts with alternative strategies for inhibition of inflammation in the treatment of CAD that prioritize broader anti-inflammatory mechanisms, including inhibition of IL1 β or IL6.⁴³

CONCLUSIONS

From the discovery of ADAR RNA editing over 30 years ago, major challenges in the field have focused on understanding the roles of RNA editing in mediating innate immune response and have been shaped by the discovery that A-to-I RNA editing can mediate dsRNA structure and dsRNA sensing by MDA5. Recent discoveries have now highlighted strong human genetic evidence to support an axis of ADAR1-dsRNA-MDA5 in the pathogenesis of numerous autoinflammatory disorders as well as CAD. From this context, there is now a rapid advancement in our understanding of this mechanism in the development of vascular disease, where inadequate ADAR1 RNA editing, with a distinct role in the vascular SMC, leads to activation of MDA5, which regulates broad transcriptional networks that control cellular phenotype, cellular fate trajectory, and disease progression. We now have strong evidence to conclude ADAR1-dsRNA-MDA5 is a causal mechanism in atherosclerotic disease, and therapeutic strategies to target RNA sensing have tremendous potential.

ARTICLE INFORMATION

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Disclosures

C.S. Weldy is a consultant for AiRNA Bio and Avidity Biosciences. T. Quertermous serves on the scientific advisory board for Amgen. J.B. Li is a cofounder of AiRNA Bio and a consultant for Risen Pharma. J.B. Li is named as inventor of a patent filed by Stanford University (WO/2023/239781), describing a method related to genetic prediction of RNA editing that relates to human prediction of double-stranded RNA burden.

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